

We shall now look at the solutions of Schrödinger's equation for the quantum harmonic oscillator. In this chapter we will focus on the one-dimensional case, which can be seen as another example of a one-dimensional potential. Unlike the infinite potential well, the potential for the harmonic oscillator is finite for all finite values of x , and only diverges when $x \rightarrow \pm\infty$. As we will see, the solution of the time-independent Schrödinger equation leads to a quantised spectrum. Stationary states correspond to quanta of energy. An important new methodology is introduced in this chapter: the time-independent Schrödinger equation will be solved using creation and annihilation operators, a technique that we will use again later in this book. Creation and annihilation operators yield a simple and physical interpretation of the energy eigenstates, and are used widely, e.g. in relativistic theories to build the space of physical states in quantum field theories. We will also solve the eigenvalue equation as a differential equation for the wave function. It is important to compare and understand the relation between these two different methods. The harmonic oscillator is ubiquitous in physics: every system close to the minimum of the potential is described by a quadratic potential, i.e. it behaves like a harmonic oscillator. The techniques and results presented in this chapter are applied widely to a variety of physical systems.

6.1 The Harmonic Oscillator in Classical Mechanics

Classically, the equation of motion for a particle of mass m , subject to a restoring force proportional to the displacement, x , is

$$m\ddot{x} = -kx, \quad (6.1)$$

where the constant of proportionality, k , is usually called the spring constant. This has oscillatory solutions of angular frequency $\omega = \sqrt{k/m}$. Energy conservation can be obtained by writing

$$\ddot{x} = v \frac{dv}{dx}, \quad (6.2)$$

and integrating the equation of motion above to give

$$\frac{1}{2}mv^2 + \frac{1}{2}kx^2 = E, \quad (6.3)$$

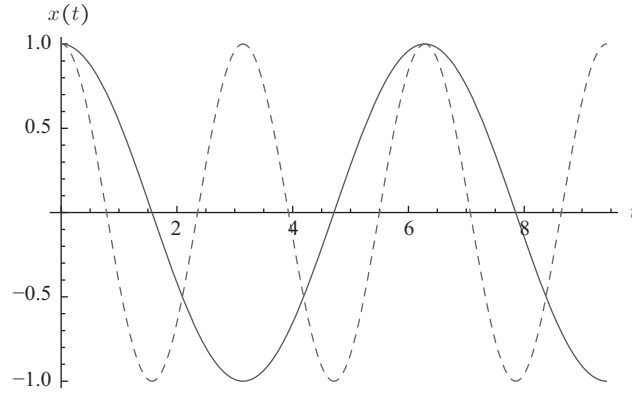


Fig. 6.1

Classical trajectory of the harmonic oscillator for $\omega = 1$ (solid curve) and $\omega = 2$ (dashed curve). The curves represent the position of the system as a function of time.

where E , the total energy of the system, is a constant. The first term on the left-hand side is the kinetic energy of the system, while the potential energy function, V , is given by

$$V(x) = \frac{1}{2}kx^2 = \frac{1}{2}m\omega^2x^2. \quad (6.4)$$

It is the quadratic dependence on the position that characterises the harmonic motion. The evolution of the position of the oscillator as a function of time $x(t)$ is shown in Fig. 6.1 for two different values of ω . The total energy of the oscillator is $E = \frac{1}{2}m\omega^2L^2$, where L is the maximum amplitude of the oscillations. The larger the amplitude of the oscillations, the larger the energy of the system. In the case depicted in Fig. 6.1, the amplitude $L = 1$ was used for both curves.

6.2 The Quantum Harmonic Oscillator

By analogy with the classical case, the quantum version of the **harmonic oscillator** is a one-dimensional system (for now), whose potential is a quadratic function of its coordinate. The time-independent Schrödinger equation is

$$\hat{H}|n\rangle = E_n|n\rangle, \quad (6.5)$$

where we denote by $|n\rangle$ the eigenstates of the Hamiltonian

$$\hat{H} = \frac{\hat{P}^2}{2m} + \frac{1}{2}m\omega^2\hat{X}^2. \quad (6.6)$$

The notation used above suggests that the spectrum is discrete and the eigenstates of the Hamiltonian are labelled by some integer n . We will see below that this is indeed the case.

Taking the scalar product of Eq. (6.5) with eigenstates of the position operator $|x\rangle$, we can recast the time-independent Schrödinger equation as a differential equation for the wave functions of the eigenstates:

$$\left[-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2} m \omega^2 x^2 \right] u_n(x) = E_n u_n(x), \quad (6.7)$$

where the wave function is defined as usual:

$$u_n(x) = \langle x | n \rangle.$$

We will discuss the detailed solution of this differential equation at the very end of the chapter. Before dwelling on the details of such derivations, we will develop an elegant algebraic method to find the eigenfunctions. For now it suffices to say that acceptable solutions only arise for certain values of the total energy, i.e. the energy is *quantised*. The **energy eigenvalues** are

$$E_n = (n + \frac{1}{2}) \hbar \omega, \quad n = 0, 1, 2, 3, \dots, \infty. \quad (6.8)$$

Note that $\hbar \omega$ has got the correct dimensions, i.e. it is an energy. The state of lowest energy, or *ground state*, is labelled $n = 0$ and has an energy $\frac{1}{2} \hbar \omega$ rather than zero. The corresponding **eigenfunctions** may be written as

$$u_n(x) = C_n \exp\{-\alpha^2 x^2/2\} H_n(\alpha x), \quad (6.9)$$

where C_n is a normalisation constant, $\alpha^2 \equiv m\omega/\hbar$ and $H_n(\alpha x)$ are polynomials of degree n , known as *Hermite polynomials*. Once again using dimensional analysis, you can show that α has dimensions of inverse length, so that the product αx is dimensionless. The Hermite polynomials satisfy the orthogonality relation

$$\int_{-\infty}^{\infty} \exp\{-s^2\} H_m(s) H_n(s) ds = 2^n \sqrt{\pi} n! \delta_{mn}, \quad (6.10)$$

implying that the energy eigenfunctions are orthogonal (orthonormal with an appropriate choice of the constant, C_n).

The first few Hermite polynomials are

$$H_0(s) = 1, \quad (6.11)$$

$$H_1(s) = 2s, \quad (6.12)$$

$$H_2(s) = 4s^2 - 2, \quad (6.13)$$

$$H_3(s) = 8s^3 - 12s. \quad (6.14)$$

Note that the polynomials, and hence the energy eigenfunctions, are alternately even and odd functions of x , i.e. they are of either even or odd parity. Acting with the parity operator introduced in Eq. (4.49), you can readily check that

$$\hat{P} u_n(x) = (-1)^n u_n(x), \quad (6.15)$$

the parity of the state u_n is $(-1)^n$. This is consistent with the discussion at the end of Chapter 5, since the potential for the harmonic oscillator is symmetric under reflections:

$$V(x) = V(-x). \quad (6.16)$$

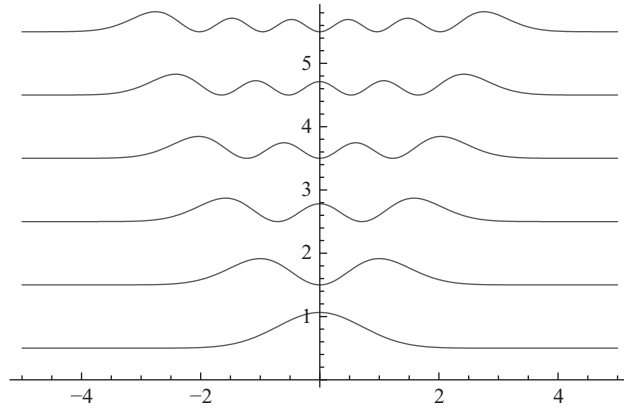


Fig. 6.2

Probability of finding the oscillator at x for the first five levels of the harmonic oscillator for $\alpha = 1$. Each curve is shifted upwards by $n + \frac{1}{2}$ to improve the readability of the plot.

The probability density to find the oscillator at a given position x is $|u_n(x)|^2$. The probability densities for the first five levels are shown in Fig. 6.2. It is clear from the shape of the probability densities in the figure that the probability of finding the system at larger values of x increases as the energy of the state is increased, i.e. for larger values of n . In order to quantify the last sentence we can compute the average position, and the fluctuations of the position, when the system is in the n th stationary state. The expectation value of the position of the quantum harmonic oscillator is given by

$$\langle x \rangle_n = \int_{-\infty}^{+\infty} dx \, x \left| C_n \exp\{-\alpha^2 x^2/2\} H_n(\alpha x) \right|^2, \quad (6.17)$$

where the suffix n is there to remind us that the expectation value is computed in the state $u_n(x)$. You can easily convince yourself that $\langle x \rangle = 0$, because the integrand is always an *odd* function of x . The vanishing of the mean value of the position is what we would have naively expected, since the system is symmetric under parity. The quantum oscillator is equally likely to be to the left or to the right of the origin at $x = 0$. Note that here we are *not* considering the time evolution of the oscillator, but rather the probability density of finding the system at position x when the state of the system is described by one of the stationary states. The average in Eq. (6.17), and the average discussed in the next paragraph, is not an average over time, it is the average over multiple measurements performed after preparing the system always in the same state u_n .

It is more instructive to compute the mean square deviation $\langle x^2 \rangle$, which measures the variance of the probability distribution of the outcome of a measurement of the position of the system. You can compute the integrals explicitly for the first few Hermite polynomials, or simply look up the answer in a table of integrals:

$$\langle x^2 \rangle_n = \left(n + \frac{1}{2} \right) \frac{1}{\alpha^2}. \quad (6.18)$$

Once again this is the quantitative expression of some intuitive understanding: the larger the energy, the larger the variance. Observations of a large value of the position of the system become more probable as the energy of the system increases, more precisely:

$$E_n = m\omega^2 \langle x^2 \rangle_n, \quad (6.19)$$

which is the quantum version of the relation between the energy and the size of the oscillations of the harmonic oscillator. As you can see in Eq. (6.19), we found a relation between observables: the energy of the system and its position. In a stationary state, measurements of the energy will yield the eigenvalue E_n with probability 1. Measurements of the position will yield a distribution of values – remember that the results of experiments are stochastic variables – with mean value equal to 0 and variance proportional to the energy of the state. This is the kind of intuition that you may want to develop in quantum mechanics, in order to make contact between the abstract formalism and physical reality.

In the following sections we shall proceed to solve the time-independent Schrödinger equation. The solution will involve a number of steps:

1. we will rewrite the Hamiltonian in a factorised form;
2. we will then introduce and discuss creation and annihilation operators;
3. finally, we will be able to construct explicitly the solutions.

6.3 Factorising the Hamiltonian

The Hamiltonian for the harmonic oscillator is

$$\hat{H} = \frac{\hat{P}^2}{2m} + \frac{1}{2}m\omega^2 \hat{X}^2. \quad (6.20)$$

Let us factor out $\hbar\omega$, and rewrite the Hamiltonian as

$$\hat{H} = \hbar\omega \left[\frac{\hat{P}^2}{2m\hbar\omega} + \frac{m\omega}{2\hbar} \hat{X}^2 \right]. \quad (6.21)$$

Checking the dimensions of the constants, you can readily verify that

$$[\hbar\omega] = \text{energy}, \quad [2m\omega\hbar] = \text{momentum}^2, \quad \left[\frac{2\hbar}{m\omega} \right] = \text{length}^2. \quad (6.22)$$

Introducing the dimensionless quantities

$$\hat{\xi} = \sqrt{\frac{m\omega}{2\hbar}} \hat{X}, \quad (6.23)$$

$$\hat{\eta} = \frac{\hat{P}}{\sqrt{2m\hbar\omega}}, \quad (6.24)$$

the Hamiltonian becomes

$$\hat{H} = \hbar\omega \left[\hat{\eta}^2 + \hat{\xi}^2 \right]. \quad (6.25)$$

The operators $\hat{\xi}$ and $\hat{\eta}$ are simply the position and the momentum operators rescaled by some real constants; therefore both of them are Hermitian. Their commutation relation can easily be computed using the canonical commutation relations

$$[\hat{\xi}, \hat{\eta}] = \frac{1}{2\hbar} [\hat{X}, \hat{P}] = \frac{i}{2}. \quad (6.26)$$

If $\hat{\xi}$ and $\hat{\eta}$ were commuting variables, we would be tempted to factorise the Hamiltonian as

$$\hat{H} = \hbar\omega (\hat{\xi} + i\hat{\eta})(\hat{\xi} - i\hat{\eta}). \quad (6.27)$$

However, we must be careful here, because the operators do not commute. So let us introduce

$$\begin{cases} \hat{a} = \hat{\xi} + i\hat{\eta}, \\ \hat{a}^\dagger = \hat{\xi} - i\hat{\eta}; \end{cases} \quad (6.28)$$

the expressions for $\hat{a}$ and $\hat{a}^\dagger$ in terms of $\hat{X}$ and $\hat{P}$ are

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}} \hat{X} + \frac{i}{\sqrt{2m\omega\hbar}} \hat{P}, \quad (6.29)$$

$$\hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \hat{X} - \frac{i}{\sqrt{2m\omega\hbar}} \hat{P}. \quad (6.30)$$

We can then compute

$$\hat{a}\hat{a}^\dagger = \hat{\xi}^2 + i[\hat{\eta}, \hat{\xi}] + \hat{\eta}^2, \quad (6.31)$$

$$\hat{a}^\dagger\hat{a} = \hat{\xi}^2 - i[\hat{\eta}, \hat{\xi}] + \hat{\eta}^2. \quad (6.32)$$

Summing the two equations above, we get a factorised expression for the Hamiltonian:

$$\hat{H} = \frac{\hbar\omega}{2} (\hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a}). \quad (6.33)$$

Subtracting the same two equations yields the commutation relation between $\hat{a}$ and $\hat{a}^\dagger$:

$$[\hat{a}, \hat{a}^\dagger] = 1. \quad (6.34)$$

This commutation relation plays an important role in the rest of this chapter. Using the commutation relation, we can derive an alternative, and more useful, expression for $\hat{H}$:

$$\hat{H} = \left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right)\hbar\omega. \quad (6.35)$$

6.4 Creation and Annihilation

We are now going to find the eigenvalues of $\hat{H}$ using the operators $\hat{a}$ and $\hat{a}^\dagger$. We begin by computing the commutators $[\hat{H}, \hat{a}]$ and $[\hat{H}, \hat{a}^\dagger]$:

$$[\hat{H}, \hat{a}] = \left[\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right)\hbar\omega, \hat{a}\right] = \hbar\omega[\hat{a}^\dagger\hat{a}, \hat{a}], \quad \text{since} \quad \left[\frac{1}{2}, \hat{a}\right] = 0. \quad (6.36)$$

Now

$$[\hat{a}^\dagger \hat{a}, \hat{a}] = [\hat{a}^\dagger, \hat{a}] \hat{a} = -\hat{a}, \quad (6.37)$$

so that we obtain

$$[\hat{H}, \hat{a}] = -\hbar\omega \hat{a}. \quad (6.38)$$

Similarly

$$[\hat{H}, \hat{a}^\dagger] = \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) \hbar\omega, \hat{a}^\dagger] = \hbar\omega [\hat{a}^\dagger \hat{a}, \hat{a}^\dagger], \quad \text{since} \quad \left[\frac{1}{2}, \hat{a}^\dagger \right] = 0, \quad (6.39)$$

and

$$[\hat{a}^\dagger \hat{a}, \hat{a}^\dagger] = \hat{a}^\dagger [\hat{a}, \hat{a}^\dagger] = \hat{a}^\dagger, \quad (6.40)$$

and therefore

$$[\hat{H}, \hat{a}^\dagger] = \hbar\omega \hat{a}^\dagger. \quad (6.41)$$

Using these relations, we can compute

$$\hat{H}(\hat{a}|n\rangle) = \hat{a}\hat{H}|n\rangle + [\hat{H}, \hat{a}]|n\rangle, \quad (6.42)$$

$$= E_n \hat{a}|n\rangle - \hbar\omega \hat{a}|n\rangle, \quad (6.43)$$

$$= (E_n - \hbar\omega)(\hat{a}|n\rangle). \quad (6.44)$$

We have found an eigenvalue equation: it states that $\hat{a}|n\rangle$ is an eigenfunction of $\hat{H}$ belonging to the eigenvalue $(E_n - \hbar\omega)$, unless $\hat{a}|n\rangle \equiv 0$. We say that the operator $\hat{a}$ is a **lowering operator**; its action on an energy eigenstate is to turn it into another energy eigenstate of lower energy. It is also called an **annihilation operator**, because it removes one quantum of energy $\hbar\omega$ from the system.

Similarly, it is straightforward to show that

$$\hat{H}\hat{a}^\dagger|n\rangle = (E_n + \hbar\omega)\hat{a}^\dagger|n\rangle, \quad (6.45)$$

which says that $\hat{a}^\dagger|n\rangle$ is an eigenfunction of $\hat{H}$ belonging to the eigenvalue $(E_n + \hbar\omega)$, unless $\hat{a}^\dagger|n\rangle \equiv 0$. We say that the operator $\hat{a}^\dagger$ is a **raising operator**; its action on an energy eigenstate is to turn it into another energy eigenstate of higher energy. It is also called a **creation operator**, because it adds one quantum of energy $\hbar\omega$ to the system.

We can summarise these results by denoting the states of energy $E_n \pm \hbar\omega$ by $|n \pm 1\rangle$ and writing

$$\hat{a}|n\rangle = c_n|n-1\rangle \quad \text{and} \quad \hat{a}^\dagger|n\rangle = d_n|n+1\rangle, \quad (6.46)$$

where c_n and d_n are constants of proportionality (NOT eigenvalues!) and

$$\hat{H}|n-1\rangle = E_{n-1}|n-1\rangle = (E_n - \hbar\omega)|n-1\rangle, \quad (6.47)$$

$$\hat{H}|n+1\rangle = E_{n+1}|n+1\rangle = (E_n + \hbar\omega)|n+1\rangle. \quad (6.48)$$

6.5 Eigensystem

6.5.1 Eigenvalues

It should be clear that repeated applications of the lowering operator, $\hat{a}$, generate states of successively lower energy *ad infinitum unless there is a state of lowest energy*; application of the operator to such a state must yield the null vector.

Does such a state exist? The answer is yes because the Hamiltonian can only have positive eigenvalues, since the minimum of the potential is $V_{\min} = 0$ as discussed in the general case in [Chapter 4](#). Alternatively, you can consider the expectation value of $\hat{H}$ in an arbitrary state $|\Psi\rangle$:

$$\langle \hat{H} \rangle = \left\langle \frac{\hat{p}^2}{2m} \right\rangle + \left\langle \frac{1}{2} m \omega^2 \hat{x}^2 \right\rangle, \quad (6.49)$$

and both terms on the right-hand side are non-negative. Thus there cannot be any states of negative energy.¹

We denote the state of lowest energy, or ground state, by $|0\rangle$. Since there cannot be a state of lower energy, we must have

$$\hat{a} |0\rangle = |\emptyset\rangle, \quad (6.50)$$

where we introduced $|\emptyset\rangle$ to denote the null vector in the space of physical states, so that it does not get confused with the ground state $|0\rangle$. Applying the Hamiltonian to this state we see that

$$\hat{H} |0\rangle = \hbar\omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) |0\rangle = \frac{1}{2} \hbar\omega |0\rangle \equiv E_0 |0\rangle. \quad (6.51)$$

Thus we have found the ground-state energy: $E_0 = \frac{1}{2} \hbar\omega$. Application of the raising operator to the ground state generates the state $|1\rangle$ with energy $E_1 = \frac{3}{2} \hbar\omega$, whilst n applications of the raising operator generate the state $|n\rangle$ with energy $(n + \frac{1}{2}) \hbar\omega$, so that

$$E_n = (n + \frac{1}{2}) \hbar\omega, \quad n = 0, 1, 2, 3, \dots, \quad (6.52)$$

which is the result we stated at the beginning of this chapter for the energy eigenvalues of the one-dimensional oscillator!

6.5.2 Normalisation of Eigenstates

Requiring that both $|n\rangle$ and $|n-1\rangle$ be normalised enables us to determine the constant of proportionality c_n . Consider

$$\langle n | \hat{a}^\dagger \hat{a} | n \rangle = c_n \langle n | \hat{a}^\dagger | n-1 \rangle \quad \text{from property of } \hat{a}, \quad (6.53)$$

$$= c_n \langle n-1 | \hat{a} | n \rangle^* \quad \text{from definition of } \dagger, \quad (6.54)$$

$$= c_n c_n^* \langle n-1 | n-1 \rangle^* \quad \text{from property of } \hat{a}, \quad (6.55)$$

$$= |c_n|^2 \quad \text{since } \langle n-1 | n-1 \rangle^* = 1. \quad (6.56)$$

¹ This reasoning is identical to what we used in the general case, earlier in this book. It is applied here to the special case of a harmonic potential.

We can evaluate the left-hand side if we note that $\hat{a}^\dagger \hat{a} = (\hat{H}/\hbar\omega) - \frac{1}{2}$, giving

$$\langle n | \hat{a}^\dagger \hat{a} | n \rangle = n \langle n | n \rangle = n. \quad (6.57)$$

Thus $|c_n|^2 = n$ and if we choose the phase so that c_n is real, we can write

$$c_n = \sqrt{n}. \quad (6.58)$$

A similar calculation shows that

$$\langle n | \hat{a} \hat{a}^\dagger | n \rangle = |d_n|^2 = (n + 1), \quad (6.59)$$

so that if we again choose the phase so that d_n is real, we obtain

$$d_n = \sqrt{n + 1}. \quad (6.60)$$

In summary then, we have

$$\hat{a} |n\rangle = \sqrt{n} |n - 1\rangle \quad \text{and} \quad \hat{a}^\dagger |n\rangle = \sqrt{n + 1} |n + 1\rangle. \quad (6.61)$$

6.5.3 Wave Functions

Finally let us show that we can reproduce the analytic expression for the eigenfunctions of the energy. The ground state is defined by the relation

$$\hat{a} |0\rangle = |\emptyset\rangle. \quad (6.62)$$

We can rewrite the equation above as a differential operator acting on the wave function of the ground state $u_0(x)$:

$$\hat{a}u_0(x) = \left[\sqrt{\frac{m\omega}{2\hbar}} \hat{X} + \frac{i}{\sqrt{2m\omega\hbar}} \hat{P} \right] u_0(x), \quad (6.63)$$

$$= \left[\sqrt{\frac{m\omega}{2\hbar}} x + \sqrt{\frac{\hbar}{2m\omega}} \frac{d}{dx} \right] u_0(x), \quad (6.64)$$

$$= 0. \quad (6.65)$$

Hence

$$\frac{d}{dx} u_0(x) = -\sqrt{\frac{2m\omega}{\hbar}} \sqrt{\frac{m\omega}{2\hbar}} x u_0(x), \quad (6.66)$$

$$= -\frac{m\omega}{\hbar} x u_0(x), \quad (6.67)$$

$$= -\alpha^2 x u_0(x), \quad (6.68)$$

where $\alpha^2 = m\omega/\hbar$. The solution of the equation above is the Gaussian that we have already seen in [Section 6.2](#):

$$u_0(x) = C_0 \exp[-\alpha^2 x^2/2]. \quad (6.69)$$

Every other eigenfunction is obtained by repeatedly applying the creation operator $\hat{a}^\dagger$ to the ground state:

$$u_n(x) = \frac{1}{\sqrt{n!}} (\hat{a}^\dagger)^n u_0(x). \quad (6.70)$$

Remember that $\hat{a}^\dagger$ is represented as a differential operator acting on wave functions.

Exercise 6.5.1 Check that you can reproduce the wave functions for the first and second excited states of the harmonic oscillator, including the correct normalisation.

6.6 Brute Force Solution

Finally let us discuss the direct solution of the time-independent Schrödinger equation cast as a differential equation for the wave function. The method can be applied more generally to second-order equations, and indeed we will use it again to find the energy spectrum of the hydrogen atom. The equation that we are trying to solve for the harmonic oscillator is

$$\left[-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2} m \omega^2 x^2 \right] \psi(x) = E \psi(x). \quad (6.71)$$

It is often useful, when solving mathematical equations, to work with dimensionless variables – after all, mathematicians don't care about metres, Joules, seconds, etc. Following our earlier convention we introduce the dimensionless variables

$$\xi = \sqrt{\frac{m\omega}{\hbar}} x, \quad \epsilon = E/(\hbar\omega), \quad \phi(\xi) = \sqrt{\frac{\hbar}{m\omega}} \psi(x). \quad (6.72)$$

Note that we have omitted a factor of $1/\sqrt{2}$ in the definition of ξ in order to get a more symmetric differential equation below, and we have rescaled the wave function by a factor $\sqrt{\frac{\hbar}{m\omega}}$, so that it is also dimensionless.² The eigenvalue equation becomes

$$\left[\frac{d^2}{d\xi^2} - (\xi^2 - 2\epsilon) \right] \phi(\xi) = 0. \quad (6.73)$$

In order to proceed, note that

$$\lim_{x \rightarrow \pm\infty} \frac{d^2}{d\xi^2} \phi(\xi) = \xi^2 \phi(\xi), \quad (6.74)$$

with corrections that are suppressed as $1/\xi^2$. The Gaussian function $e^{-\xi^2/2}$ satisfies the same asymptotic behaviour, albeit with a different subleading correction. Therefore we are going to look for a solution of the form

$$\phi(\xi) = h(\xi) e^{-\xi^2/2}. \quad (6.75)$$

² Remember that the wave function of a quantum system on a line has dimensions of $\text{length}^{-1/2}$, so that its modulus squared is a probability density.

Substituting this ansatz in Eq. (6.73) yields an equation for $h(\xi)$:

$$\left[\frac{d^2}{d\xi^2} - 2\xi \frac{d}{d\xi} + (2\epsilon - 1) \right] h(\xi) = 0. \quad (6.76)$$

We are going to look for a solution by expanding $h(\xi)$ in a Taylor series around $\xi = 0$:

$$h(\xi) = \sum_{m=0}^{\infty} a_{2m} \xi^{2m+p}. \quad (6.77)$$

In this expansion we assume $a_0 \neq 0$, and we used the fact that the potential is symmetric under parity and therefore we can look for solutions that are either even or odd in ξ . In Eq. (6.77), p even (respectively odd) corresponds to an even (respectively odd) solution.

Equation (6.76) leads to a recursion relation for the coefficients of the Taylor expansion:

$$(2m + p + 2)(2m + p + 1)a_{2m+2} = (4m + 2p - 2\epsilon + 1)a_{2m}, \quad (6.78)$$

while for the lowest power of ξ we have

$$p(p - 1)a_0 \xi^{p-2} = 0, \quad (6.79)$$

and therefore $p = 0, 1$, since a_0 does not vanish. The recursion relation implies that, for $m \rightarrow \infty$:

$$\frac{a_{2m+2}}{a_{2m}} \sim \frac{1}{m}. \quad (6.80)$$

In turn, Eq. (6.80) can be used to show that $h(\xi)$ grows like $e^{\lambda \xi^2}$ with $0 < \lambda < 1$, and therefore the divergence of $h(\xi)$ can overcome the Gaussian that we factored out in the ansatz: $\phi(\xi)$ is unbounded for $\xi \rightarrow \pm\infty$ and the solution is not normalisable.

The only way we can have a normalisable solution is if there exists a value m_0 such that

$$(4m_0 + 2p - 2\epsilon + 1) = 0. \quad (6.81)$$

In this case $a_{2m} = 0$ for all $m > m_0$, and $h(\xi)$ is a simple polynomial. Setting $2m_0 + p = n$, the quantisation condition, Eq. (6.81), can be written

$$\epsilon = \epsilon_n = n + \frac{1}{2}, \quad (6.82)$$

which yields the energy levels that we found above using creation and annihilation operators.

Summary

- We started this chapter with a brief summary of harmonic motion in classical mechanics, which we want to contrast with the quantum solution. We should keep in mind in particular the relation between the size of the oscillations and the energy of the classical system, and compare it later on with the quantum result. (6.1)

- Before discussing the different techniques to solve the eigenvalue problem for the Hamiltonian, we focus on the statement of the problem, the form of the eigenvalue equation and its solutions. In the first place the solutions are just given and their properties are discussed. (6.2)
- We develop the algebraic solution of the problem, starting from finding a factorised expression for the Hamiltonian. (6.3)
- From the factorised expression we can then define creation and annihilation operators and discuss their properties. (6.4)
- Using the creation and annihilation operators, we proceed to the construction of the eigenvalues and eigenvectors. A simple calculation allows us to determine the normalisation of the eigenstates. (6.5)
- Having found the normalised eigenvectors, we derive the corresponding wave functions, by expanding the eigenvectors in a basis made up of eigenvectors of the position operator. (6.5.3)
- Finally we compute directly the solutions of the eigenvalue equation as a differential equation, and show that we recover the quantisation condition and the same solutions that were found using creation and annihilation operators. (6.6)

Problems

- 6.1** Write the position operator $\hat{X}$ as a function of the raising and lowering operators $\hat{a}$ and $\hat{a}^\dagger$.

Deduce the matrix element of $\hat{X}$ between two eigenstates of the Hamiltonian: $\langle n | \hat{X} | m \rangle$.

Repeat the same computations for the momentum operator $\hat{P}$.

- 6.2** Consider a one-dimensional harmonic oscillator, consisting of a particle of mass m in a quadratic potential characterised by the frequency ω . Assume that the particle has a charge q and is placed in a uniform electric field $\mathcal{E}$ in the x direction. The Hamiltonian can be written as

$$\hat{H}(\mathcal{E}) = \frac{\hat{P}^2}{2m} + \frac{1}{2}m\omega^2 \hat{X}^2 - q\mathcal{E}\hat{X}.$$

Write the eigenvalue equation for $\hat{H}(\mathcal{E})$.

Replacing the variable x by a new variable t :

$$t = x - \frac{q\mathcal{E}}{m\omega^2},$$

find the eigenvalues E'_n and the eigenfunctions $u'_n(x)$ that solve the equation above.

- 6.3** The contribution of the electron to the electric dipole moment is given by the matrix element of the operator:

$$\hat{D} = q\hat{X}.$$

In the absence of an electric field, show that

$$\langle \hat{D} \rangle = q \langle n | \hat{X} | n \rangle = 0.$$

Let us assume that the electric field is turned on. Compute the mean dipole moment:

$$\langle \hat{D} \rangle' = q \langle n' | \hat{X} | n' \rangle,$$

where $|n'\rangle$ are the kets associated with $u'_n(x)$, the wave functions found in the presence of an electric field in the question above.

Discuss the physical interpretation of this result.

- 6.4** Let us consider a one-dimensional harmonic oscillator. By recalling the action of the lowering operator a on the ground state, find the first-order differential equation that is satisfied by the wave function $u_0(x)$.

Using the creation operator, build the wave function for the first two excited states.

- 6.5** Let us consider the one-dimensional harmonic oscillator. Express $\hat{X}^2$ and $\hat{P}^2$ in terms of a and $a^\dagger$, and show that for an energy eigenstate:

$$\langle n | \frac{\hat{P}^2}{2m} | n \rangle = \langle n | \frac{1}{2} m \omega^2 \hat{X}^2 | n \rangle = \frac{1}{2} \langle n | \hat{H} | n \rangle.$$

Show that this result remains true when the oscillator is not in an eigenstate of the Hamiltonian, provided a time average over one classical oscillator period is taken. The time average is defined as

$$\bar{f} = \frac{1}{T} \int_0^T dt f(t).$$

- 6.6** Consider a one-dimensional harmonic oscillator defined by the potential

$$V(x) = \frac{1}{2} m \omega^2 x^2.$$

At $t = 0$ the state of the oscillator is

$$\Psi(x, 0) = \sum_{n=0}^{\infty} c_n u_n(x),$$

where $u_n(x)$ are the eigenfunctions of the Hamiltonian.

Write down the equations satisfied by the wave functions $u_n(x)$. Recall the possible values of the energy E_n , and the corresponding eigenfunctions.

Write the wave function describing the system at time t .

Deduce the mean value of any observable A at time t :

$$\langle \Psi(t) | \hat{A} | \Psi(t) \rangle$$

as a function of the matrix elements

$$A_{mn} = \langle u_m | \hat{A} | u_n \rangle.$$

Compare with the time dependence of the mean value of A in a stationary state, i.e. in the case where only one of the coefficients c_n is different from zero.

Let us now consider the case where $c_0 = \cos \eta$, $c_1 = \sin \eta$ and $c_n = 0$ for $n > 1$. You should use this initial state in all subsequent questions.

What are the possible outcomes of a measurement of the energy? What are the probabilities of each outcome?

Consider the observables X and P . Using the explicit expressions for the eigenfunctions $u_0(x)$ and $u_1(x)$ in Eq. (6.9), determine which are the matrix elements X_{mn} and P_{mn} that do not vanish for $m, n = 0, 1$. (You don't need to compute the integrals, just identify those that do not vanish!)

Deduce that the mean values $\langle \Psi(t) | \hat{X} | \Psi(t) \rangle$ and $\langle \Psi(t) | \hat{P} | \Psi(t) \rangle$ are sinusoidal functions of time with angular frequency ω .

Verify that this result is consistent with Ehrenfest's theorem.

6.7 Consider a one-dimensional harmonic oscillator, of mass m and frequency ω . At time $t_0 = 0$, the system is in the state $|\psi(0)\rangle$ characterised by the following properties:

- (a) a measurement of the energy yields $E_1 = 3/2\hbar\omega$ with probability $3/4$, and $E_2 = 5/2\hbar\omega$ with probability $1/4$;
- (b) the mean value of the position in the state $|\psi(0)\rangle$ vanishes;
- (c) the mean value of the momentum is positive.

The eigenstates of the Hamiltonian are identified by the kets $|n\rangle$:

$$\hat{H}|n\rangle = E_n|n\rangle.$$

Recall the possible values for the energy of the harmonic oscillator, and identify E_1 and E_2 . Explain why the state of the system can be written as

$$|\psi(0)\rangle = \frac{1}{2} \left[\sqrt{3}e^{i\phi_1}|1\rangle + e^{i\phi_2}|2\rangle \right].$$

Write down the mean value of $\hat{X}$ at time 0 as a function of ϕ_1 and ϕ_2 . You may use the following result:

$$\langle 1 | \hat{X} | 2 \rangle = \sqrt{\frac{\hbar}{m\omega}}.$$

Deduce a condition on ϕ_1 and ϕ_2 .

Starting from the matrix element of $[\hat{P}, \hat{H}]$, and using the canonical commutation relation between $\hat{X}$ and $\hat{P}$, show that

$$\langle 1 | \hat{P} | 2 \rangle = -im\omega \langle 1 | \hat{X} | 2 \rangle.$$

Compute the mean value of the momentum at $t = 0$:

$$\langle \psi(0) | \hat{P} | \psi(0) \rangle.$$

Deduce a second relation between ϕ_1 and ϕ_2 .

Find the solutions for ϕ_1 and ϕ_2 , and determine the state of the system at time t . What is the probability of finding E_2 if the energy is measured at time t ? What is the mean value of the position at time t ?